Anticoagulant reversal for intracranial hemorrhage (ICH) and other life threatening bleeds

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Anticoagulant	Reversal Agent	Comments/Alternatives
Warfarin – Intracranial hemorrhage (ICH)	Vitamin K 10 mg IV over 15-30 min **AND** Prothrombin Complex Concentrates (PCC) • INR 1.7 - 3.9: 25 units/kg (max 2500 units) • INR 4 – 6: 35 units/kg (max 3500 units) • INR greater than 6: 50 units/kg (max 5000 units)	FFP is not recommended with use of PCC unless otherwise indicated Use aPCC [FEIBA] in patients with a documented heparin allergy 500 units for INR <5 1000 units for INR ≥5
Warfarin – non-ICH bleeding	Vitamin K 10 mg IV over 15 – 30 min **AND** Fixed-dose Prothrombin Complex Concentrates (PCC) 1000 units	FFP is not recommended with use of PCC unless otherwise indicated Use aPCC [FEIBA] in patients with a documented heparin allergy 500 units for INR <5 1000 units for INR ≥5
Rivaroxaban	Send STAT UFH level (representative of anti-Xa activity). If UFH < 0.6, therapeutic anticoagulation is not likely; consider no reversal. Prothrombin Complex Concentrates (PCC) 50 units/kg IV once (max dose 5000 units as rounded to nearest vial size) If UFH level <0.6 and reversal is still desired, consider lower dose of PCC 25 units/kg **Do not recheck UFH level after PCC as value will not change and is not indicative of clinical hemostasis.	Do not re-dose PCC unless clinical failure identified Alternative if documented heparin allergy (i.e., cannot receive PCC): • Factor VIIa 45-90 mcg/kg (round to nearest 1 mg) OR • aPCC [FEIBA] 50 units/kg
Apixaban	Send STAT UFH level (representative of anti-Xa activity). If UFH < 0.6, therapeutic anticoagulation is not likely; consider no reversal. Prothrombin Complex Concentrates (PCC) 50 units/kg IV once (max dose 5000 units as rounded to nearest vial size) If UFH level <0.6 and reversal is still desired, consider lower dose of PCC 25 units/kg **Do not recheck UFH level after PCC as value will not change and is not indicative of clinical hemostasis.	Do not redose PCC unless clinical failure identified Alternative if documented heparin allergy (i.e., cannot receive PCC): • Factor VIIa 45-90 mcg/kg (round to nearest 1 mg) OR • aPCC [FEIBA] 50 units/kg

Edoxaban	Activated oral charcoal if ingestion occurred within 3 hours Prothrombin Complex Concentrates (PCC) 50 units/kg IV once (max dose 5000 units as rounded to nearest vial size)	Do not redose PCC unless clinical failure identified Alternative if documented heparin allergy (i.e., cannot receive PCC): • Factor VIIa 45-90 mcg/kg (round to nearest 1 mg) OR • aPCC [FEIBA] 50 units/kg
<u>Dabigatran</u>	Note: If thrombin time and aPTT are normal there is likely no anticoagulant effect and "reversal" is not necessary Activated oral charcoal if ingestion occurred within 2 hours Idarucizumab 5 gm IV once	Consider HD, especially if renal impairment If no clinical response to idarucizumab, consider Prothrombin Complex Concentrates (PCC) 50 units/kg (max dose 5000 units as rounded to nearest vial size) If patient has heparin allergy, aPCC [FEIBA] 50 units/kg may be utilized
Unfractionated heparin (UFH)	1 mg Protamine IV for each 100 units of UFH. Infusion should not exceed 5 mg/min due to risk of anaphylaxis Maximum single dose of Protamine is 50 mg, but may be repeated	Consider timing and total amount of UFH administered over the last 3 hours in calculation (see subsection below.) Increased risk of hypersensitivity reaction/anaphylaxis in patients with a fish allergy or prior protamine exposure
<u>Fondaparinux (Arixtra®)</u>	Prothrombin Complex Concentrates (PCC) 50 units/kg IV once	Alternative if documented heparin allergy (i.e., cannot receive PCC): • Factor VIIa 45-90 mcg/kg (round to nearest 1 mg) OR • aPCC [FEIBA] 50 units/kg May repeat Factor VII in 2 hours if bleeding recurs
Low molecular weight heparin(LMWH)	Enoxaparin - 1 mg protamine IV for each 1 mg of enoxaparin administered in the last 8 hours Dalteparin or Tinzaparin - 1 mg protamine IV for each 100 units of dalteparin or tinzaparin administered in last 8 hours For either regimen, may consider Factor VIIa 45-90 mcg/kg IV (rounded to nearest 1 mg) for life threatening intractable bleed	Protamine only partially reverses the activity of LMWH Note differences in dosing between enoxaparin and other LMWHs May administer 0.5 mg protamine for every 1 mg enoxaparin (or 100 units of dalteparin) if LMWH dose administered 8-12 hours prior
<u>Alteplase/Tenecteplase</u>	Cryoprecipitate 10 units IV AND Tranexamic acid 1 gm IV once now Obtain TEG and fibrinogen level 2 hours post-reversal to assess need for further therapy	If bleeding is refractory or life threatening, consider aminocaproic acid 0.1 g/kg bolus in addition to cryoprecipitate and tranexamic acid

Enteric Coated Aspirin	Best supportive care (based on PATCH trial) OR 1-unit single donor platelet if a surgical procedure is scheduled or bleeding is refractory	May consider repeat 1 single donor unit if PFA-100 (orderable in EPIC) is not normalized For refractory bleeding, may consider: • DDAVP 0.3 – 0.4 mcg/kg IV OR • Factor VIIa 45 – 90 mcg/kg (round to nearest 1 mg) IV
Clopidogrel	Best supportive care (based on PATCH trial) OR 2 units of single donor platelets if a surgical procedure is scheduled or bleeding is refractory	Consider repeating 2 units of single donor platelets every 12 hours until the PFA-100 and Verify Now (Plavix inhibition <20% or PRU over 240) assays are normal. Both tests are orderable in EPIC For refractory bleeding, may consider: • DDAVP 0.3 – 0.4mcg/kg IV OR • Factor VIIa 45 – 90 mcg/kg (round to nearest 1 mg) IV

* Please note these recommendations only apply to life-threatening situations where emergent reversal of anticoagulation is deemed necessary by the treating service. Most of the data comes from cases of ICH and applicability to other situations is less well studied. Adult Hematology consultation should be considered.

Examples of life-threatening traumatic hemorrhage may include penetrating injury to head, neck and torso, severe blunt trauma requiring PRBC transfusion, base deficit >6, lactic acid >3, severe hypotension with SBP<90, obvious external hemorrhage from the scalp, face, neck, torso, extremities.